Practical No 06

A) Create a simple website for a College Information Portal and convert it into a Progressive Application.

Project Name: CollegePortalPWA

Project Type: MVC Application

Program Code:

I.Filename: Index.cshtml

```
<div class="text-center">
    <h1>College Information Portal</h1>
    <h3>Welcome to xyz College</h3>
    <p>Courses Offered:</p>
    <ul>
        <li>BSC IT</li>
        <li>BSC CS</li>
        <li>BCA</li>
    </ul>
    <p>Contact: 9818763456</p>
</div>
```

II.Filename: manifest.json

```
{
  "name": "College Information Portal",
  "short_name": "CollegePortal",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

III.Filename: service-worker.js

```
self.addEventListener('install', function (event) {
    console.log('Service Worker Installed');
});

self.addEventListener('fetch', function (event) {
    console.log('Fetching:', event.request.url);
});
```

IV.Filename: Layout

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>@ViewData["Title"] - CollegePortalPWA3</title>
    <script type="importmap"></script>
    <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
    <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
    <link rel="stylesheet" href="~/CollegePortalPWA3.styles.css" asp-append-version="true" />
    <link rel="manifest" href="~/manifest.json" />
</head>
<body>
    <header>
        <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white border-bottom
box-shadow mb-3">
            <div class="container-fluid">
                <a class="navbar-brand" asp-area="" asp-controller="Home" asp-
action="Index">CollegePortalPWA3</a>
                <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target=".navbar-collapse" aria-controls="navbarSupportedContent" aria-expanded="false" aria-
label="Toggle navigation">
                    <span class="navbar-toggler-icon"></span>
                </button>
                <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
                    <ul class="navbar-nav flex-grow-1">
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Index">Home</a>
                        </li>
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
                        </li>
                    </ul>
                </div>
            </div>
        </nav>
    </header>

    <div class="container">
        <main role="main" class="pb-3">
            @RenderBody()
        </main>
    </div>

    <footer class="border-top footer text-muted">
        <div class="container">
            &copy; 2026 - CollegePortalPWA3 - <a asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
        </div>
    </footer>

    <script src="~/lib/jquery/dist/jquery.min.js"></script>
    <script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
    <script src="~/js/site.js" asp-append-version="true"></script>
    @await RenderSectionAsync("Scripts", required: false)

    <script>
        if ('serviceWorker' in navigator) {
            navigator.serviceWorker.register('/service-worker.js')
                .then((registration) => {
                    console.log('Service Worker Registered!', registration);
                })
                .catch((error) => {
                    console.log('Service Worker Registration Failed:', error);
                });
```

```
        }
    </script>
</body>
</html>
```

V.Filename: Program.cs

```
Adding a line – app.UseStaticFiles();
```

B) Create a PWA that displays a custom offline page when the user is not connected to the internet.

Project Name: OfflinePWA

Project Type: MVC Application

Program Code:

I.Filename: index.cshtml

```
<div class="text-center">
    <h1>College Information Portal</h1>
    <h3>Welcome to XYZ College</h3>
    <p>This website supports Offline Mode.</p>
</div>
```

II.Filename: offline.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Offline</title>
</head>
<body>
    <h1>No Internet Connection</h1>
    <h3>Please check your internet connection.</h3>
</body>
</html>
```

III.Filename: manifest.json

```
{
  "name": "Offline PWA",
  "short_name": "OfflinePWA",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "/icons/icon1.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "/icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

IV.Filename: Service-Worker

```
const CACHE_NAME = "offline-cache-v1";
const urlsToCache = [
```

```
    "/",
    "/offline.html"
];

self.addEventListener("install", event => {
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(cache => {
                return cache.addAll(urlsToCache);
            })
    );
});

self.addEventListener("fetch", event => {
    event.respondWith(
        fetch(event.request)
            .catch(() => {
                return caches.match("/offline.html");
            })
    );
});
```

V.Filename: Layout

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>@ViewData["Title"] - OfflinePWA</title>
    <script type="importmap"></script>
    <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
    <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
    <link rel="stylesheet" href="~/OfflinePWA.styles.css" asp-append-version="true" />
    <link rel="manifest" href="~/manifest.json" />
</head>
<body>
    <header>
        <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white border-
bottom box-shadow mb-3">
            <div class="container-fluid">
                <a class="navbar-brand" asp-area="" asp-controller="Home" asp-
action="Index">OfflinePWA</a>
                <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target=".navbar-collapse" aria-controls="navbarSupportedContent" aria-expanded="false" aria-
label="Toggle navigation">
                    <span class="navbar-toggler-icon"></span>
                </button>
                <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
                    <ul class="navbar-nav flex-grow-1">
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Index">Home</a>
                        </li>
                        <li class="nav-item">
                            <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
                        </li>
                    </ul>
                </div>
            </div>
        </nav>
    </header>

    <div class="container">
        <main role="main" class="pb-3">
            @RenderBody()
        </main>
```

```
    </div>

    <footer class="border-top footer text-muted">
        <div class="container">
            &copy; 2026 - OfflinePWA - <a asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
        </div>
    </footer>

    <script src="~/lib/jquery/dist/jquery.min.js"></script>
    <script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
    <script src="~/js/site.js" asp-append-version="true"></script>
    @await RenderSectionAsync("Scripts", required: false)

    <script>
        if ('serviceWorker' in navigator) {
            navigator.serviceWorker.register('/service-worker.js')
                .then(() => console.log('Service Worker Registered'));
        }
    </script>
</body>
</html>
```